

MAYURESH PRAMOD PANDEY

San Jose, CA | mayurp.pandey@gmail.com | +1 (669) 340-6006 | linkedin.com/in/mayureshpp | github.com/mayu99

EDUCATION

Master of Science, Data Analytics

Aug 2024 – May 2026

San Jose State University, CA, USA

Courses – Mathematics, Big Data, Data Visualization, Machine Learning, Deep Learning, Data Warehouse and Pipeline, Distributed Systems for Data Engineering, Generative AI Applications.

Bachelor of Engineering, Information Technology

July 2017 - June 2021

University of Mumbai, Maharashtra, India

Courses – Data Structure and Algorithm, DBMS, Data Mining, Gen AI, Machine Learning, AI Algorithms.

SKILLS

- **Languages & Tools:** SQL (joins, group by, calculated columns), Python (pandas, scikit-learn), R, Excel (VLOOKUP, PivotTables, VBA), Java
- **Visualization:** Tableau, Power BI, Power Query, Matplotlib
- **Data & ML:** Data Cleaning, Statistical Analysis, Time Series Forecasting, PySpark
- **Healthcare & Compliance:** Experience with healthcare prediction projects, knowledge of HIPAA standards, ICD-10 familiarity through research projects
- **Cloud & DevOps:** AWS, Azure, Jenkins, Docker, Git

RELEVANT EXPERIENCE

Data Engineer | Om Shanti Construction Pvt. Ltd., Mumbai, India

Jan 2023 - June 2024

(Tech Stack: - Node.js, Python, MySQL, AWS, Docker, REST APIs, Tableau.)

- Developed and maintained web and mobile applications for real estate listings, searches, and bookings, while migrating platforms to AWS for improved scalability and security.
- Automated CI/CD pipelines using Docker, Jenkins, and AWS, reducing deployment time and enhancing code reliability.
- Built data analytics tools for real-time forecasting and integrated them into Tableau for business intelligence and decision-making.
- Conducted code reviews, mentored junior developers, and collaborated with cross-functional teams, improving code quality by **40%** and reducing bug reports by **25%**.

Decision Scientist | Mu Sigma, Bangalore, India

Aug 2021 - Dec 2022

(Tech Stack: - PySpark, Hive, DBFS, Azure Data Lake, Azure Data Bricks, Azure Blob, Azure Key Vaults, Azure DevOps.)

- Developed an automated forecasting tool using Statistical Modelling and Machine Learning to enhance supply and demand planning across multiple regions (Spain, Portugal, and Poland).
- Built ML models for order and shipment forecasts using time series algorithms like ARIMA, Prophet, and Holt-Winters, and designed data pipelines with Azure Data Factory to handle large-scale data processing.
- Created data assets for an automated app on React using ETL frameworks.

PROJECTS AND PAPERS

Mental Health Prediction for Juveniles using Machine Learning Techniques

April 2021

- Developed a website and mobile app using data from a mental health research institute and collaborating with psychiatrists to create a predictive model for early detection of mental illness in juveniles, reducing the initial screening time by 50%.
- Technologies used: Spark, Django, Python, SQL, React, Node.js, Machine Learning algorithm.
- Authored and published a paper on the same in the 4th International Conference on Advances in Science and Technology-Elsevier in March-April 2021, SSRN ID: 3867291.

LLM-Based Context Retrieval and Cost Optimization (SjSU)

Dec 2024

- Designed a modular RAG pipeline using LangChain, OpenAI, and FAISS for legal document summarization.
- Reduced token usage by 35% using dynamic chunking and semantic context filtering in queries.
- Compared latency and accuracy tradeoffs across Claude, Gemini, and Mistral models in production-like scenarios.

Signova: Indian Sign Language Translation Platform

June 2024

- Built an ISL translation tool using Python (Flask), achieving 99.98% classification accuracy.
- Integrated a learning module to aid users in mastering ISL gestures.

Identity Resolution in Social Network using Recommender System

June 2021

Journal of AI and Systems, ISSN: 2642-2859

- Merged duplicate profiles and improved recommendation accuracy using hybrid algorithms.
- Tools: MongoDB, Python, Graph databases

COURSES AND CERTIFICATIONS

- IBM Full Stack Cloud Developer Professional Certificate (Coursera)
- Meta Front-End Developer Professional Certificate (Coursera)
- Machine Learning Specialization (Coursera)
- Solution Architect (AWS)